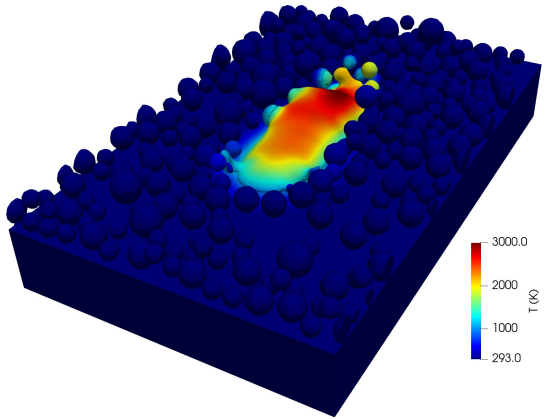
Tutorial for Process optimization for IN718:
A Single bead analysis



Access Full AM PravaH Tutorials

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Melt pool dynamics prediction using AM PravaH macroscale module



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Simulation details

Process	L-PBF
Alloy	IN718
Power (W)	285
Scan speed (m/s)	0.96
Powder morphology	Spherical
Powder size distribution (µm)	28-48
Spot diameter (µm)	100

Process	L-PBF
Layer thickness (µm)	70
Substrate height (µm)	200
Chamber pressure (Pa)	100000
Preheat temperature (K)	353
Fresnel coeff (solid-liquid)	0.5
Absorption coeff (liquid)	25000

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Case: single track - LPBF

Aim: To investigate the effect of laser power on melt pool geometry and experimental melt pool shape in the LPBF process using AM PravaH®.

Process parameters

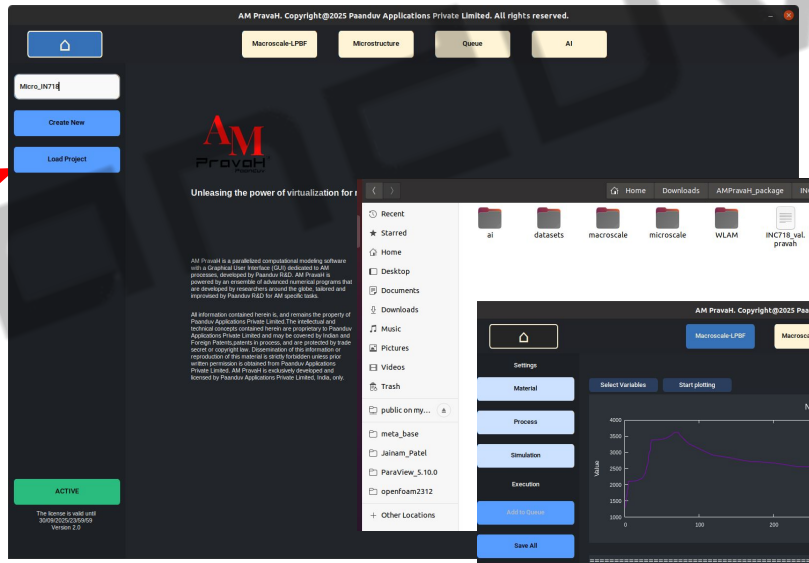
Alloy	Spot diameter (m)	Laser power (W)	Scan speed (mm/min)
IN718	10E-5	285	57600

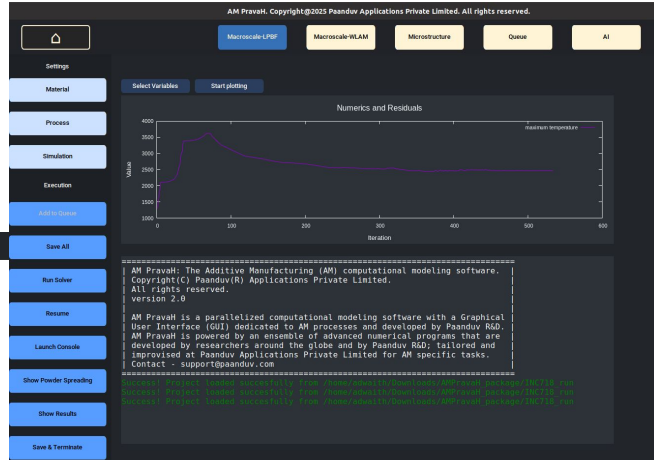
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Hands-on-Orientation on the macroscale module of AM PravaH® 2.0 Graphical User Interface (GUI)

- Pick a desired project file name,

  Let's say INC718_run in our case
- Click on the **Create New** tab below
- Save it to a desired location





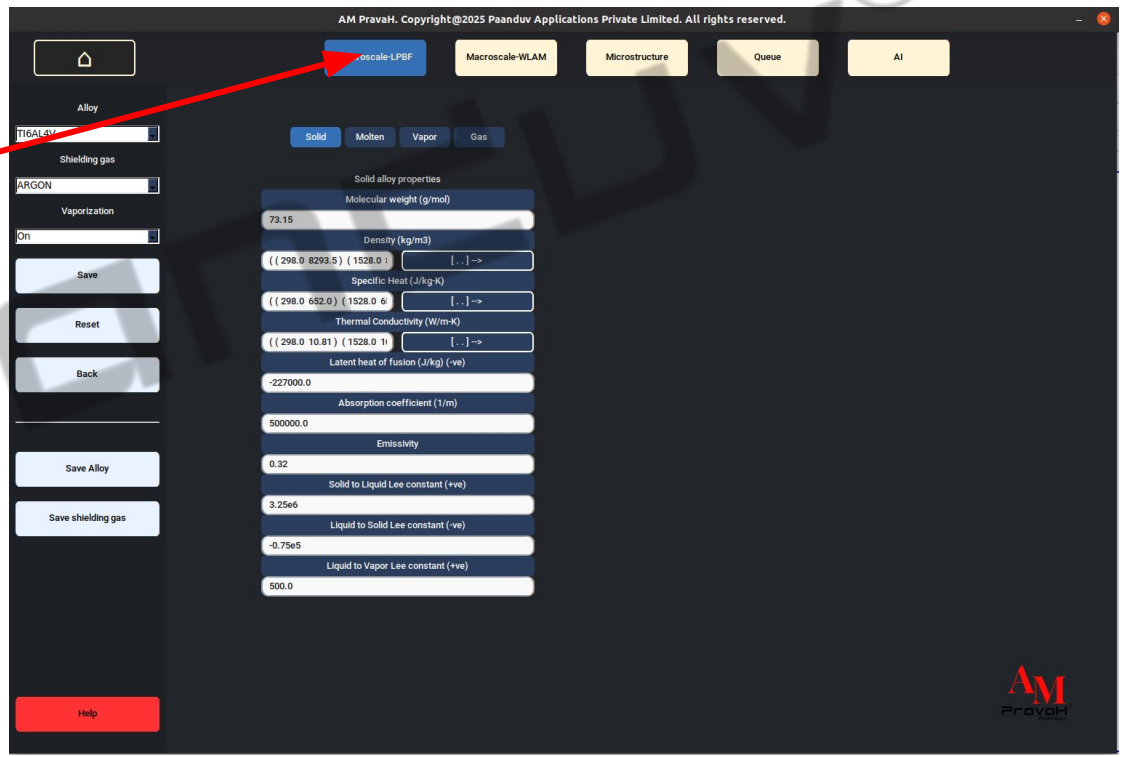
If the file is already created,

Go to **Load Project** tab and open the project (**INC718_val.pravah**) file.

The file is loaded successfully

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- Opent the AM PravaH GUI
- Choose the **Macrostructure** tab
- Select **Settings** tab from the left panel to explore further



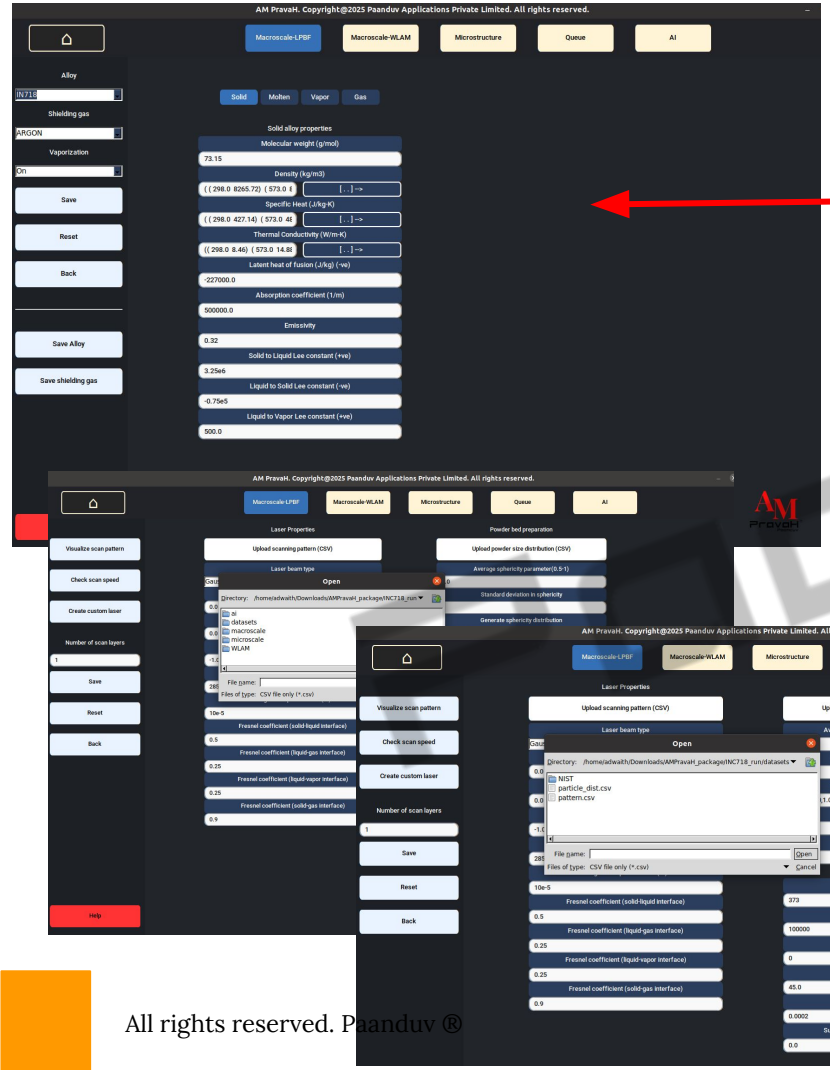
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In the Settings tab

- Select the desired alloy from the left panel
- A drop-down list of different alloys will appear
- Scroll down and select the alloy, in our case choose IN 718.
- Input the alloy properties, import scan pattern file, process parameters, simulation parameters and parallelization scheme
- The fidelity of the simulation is controlled by the sliding the Accuracy, and Performance.



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Thermophysical properties are given as inputs:

- Solidus temperature (in K),
- Thermal Diffusivity (in m^2/s)
- Thermal conductivity (in W/m-K) (Based on literature or CALPHAD)

Input process parameters:

- Scan strategy or pattern file (import as a .csv file)
- Bed preheat temperature (in K)
- Laser Scan speed (in m/s)
- Laser power (in W)
- Spot diameter (D4 sigma) (in m)
- Powder layer thickness (in m)
- Substrate height (in m)
- Heat absorption fraction is an iterative parameter.

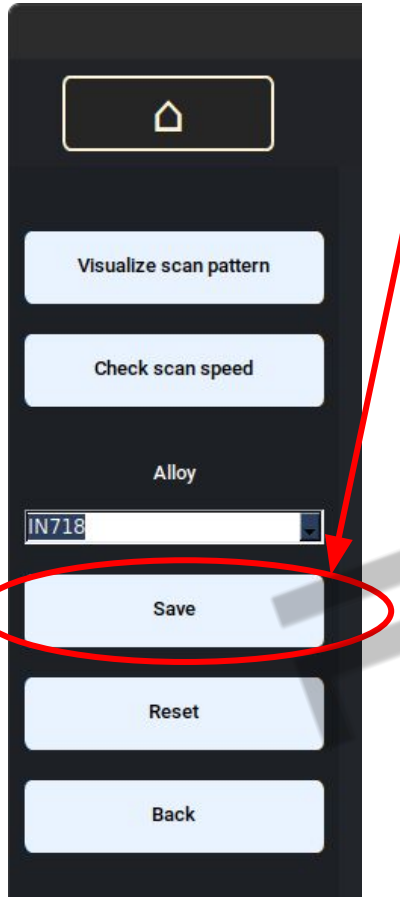
All rights reserved. Paanduv ®



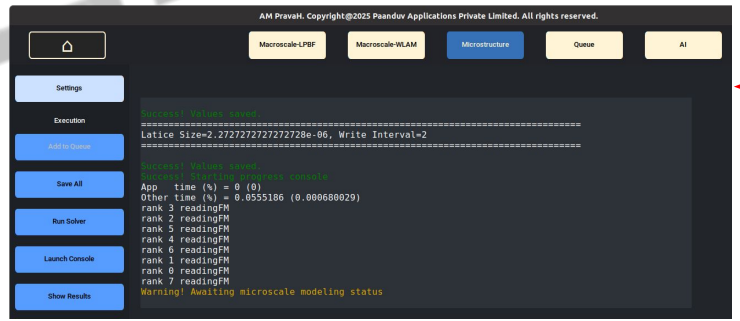
Processing efficacy: (Controls computation time)

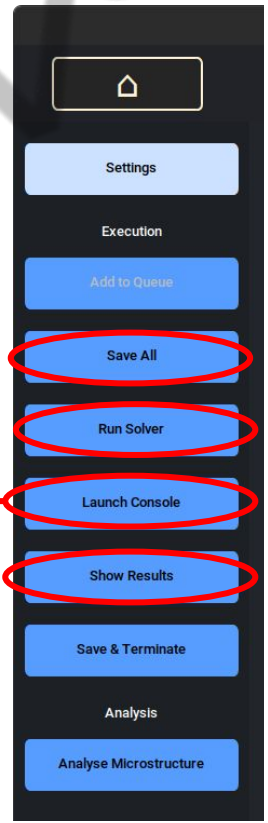
- Accuracy and performance determines resolution in the evolved structure. Computation cell size gets smaller at higher accuracies.
- Data saving frequency determines the time step.
- Parallelisation divides the domain in multiple processor cores.

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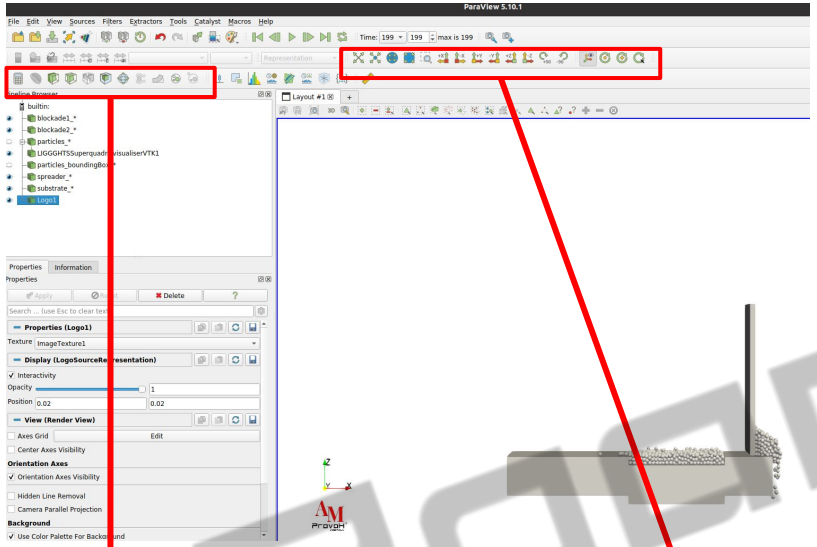


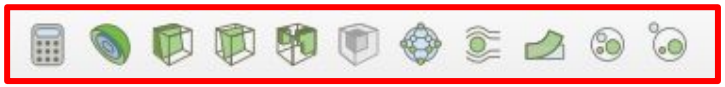
- Save all the parameters & preferences
- Go back to the project window, and click on the **Save All** tab in the left panel, to finalize the settings for a particular project.
- Click on the **Run solver** tab, to run simulations
- Click on the **Launch Console** tab. It helps in checking the status of the simulation.
- Press on **Show Results** to go to Paraview.





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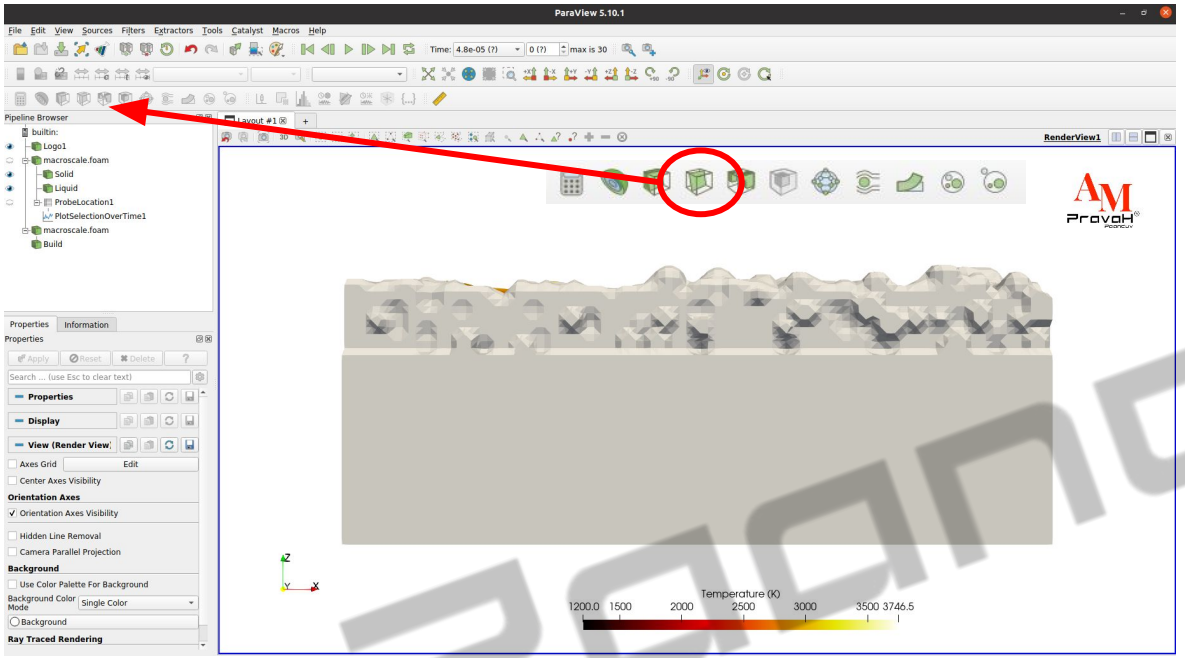
Common toolbox



Camera control toolbox

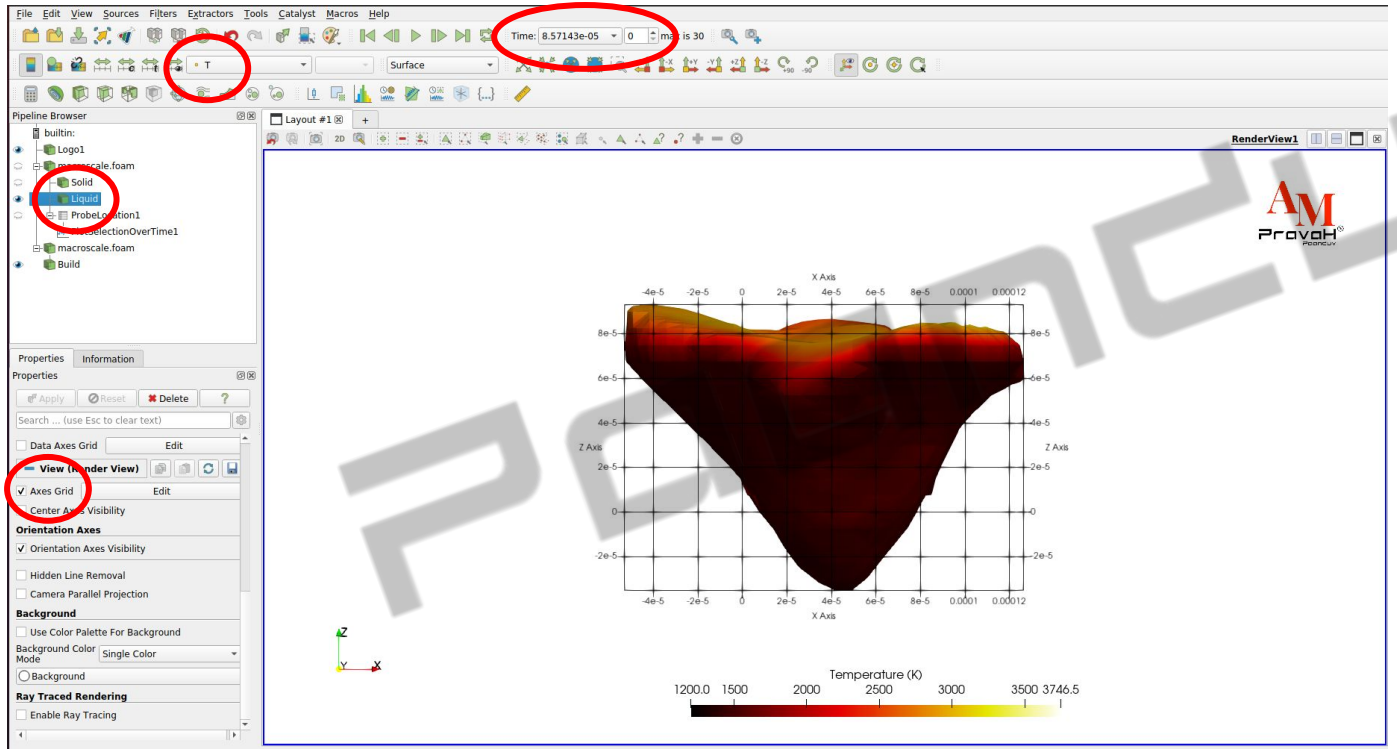
- The Paraview creates and renders the simulation
- Use the different features like clip & slice to view different sections of the melt pool
- Important tool boxes include, common tool box, camera control, etc.
- Different views will be listed in the Pipeline Browser (located at the left panel) contains all the views and clips.
- Show Powder Spreading can be used to check the spreading of metal powder over the substrate

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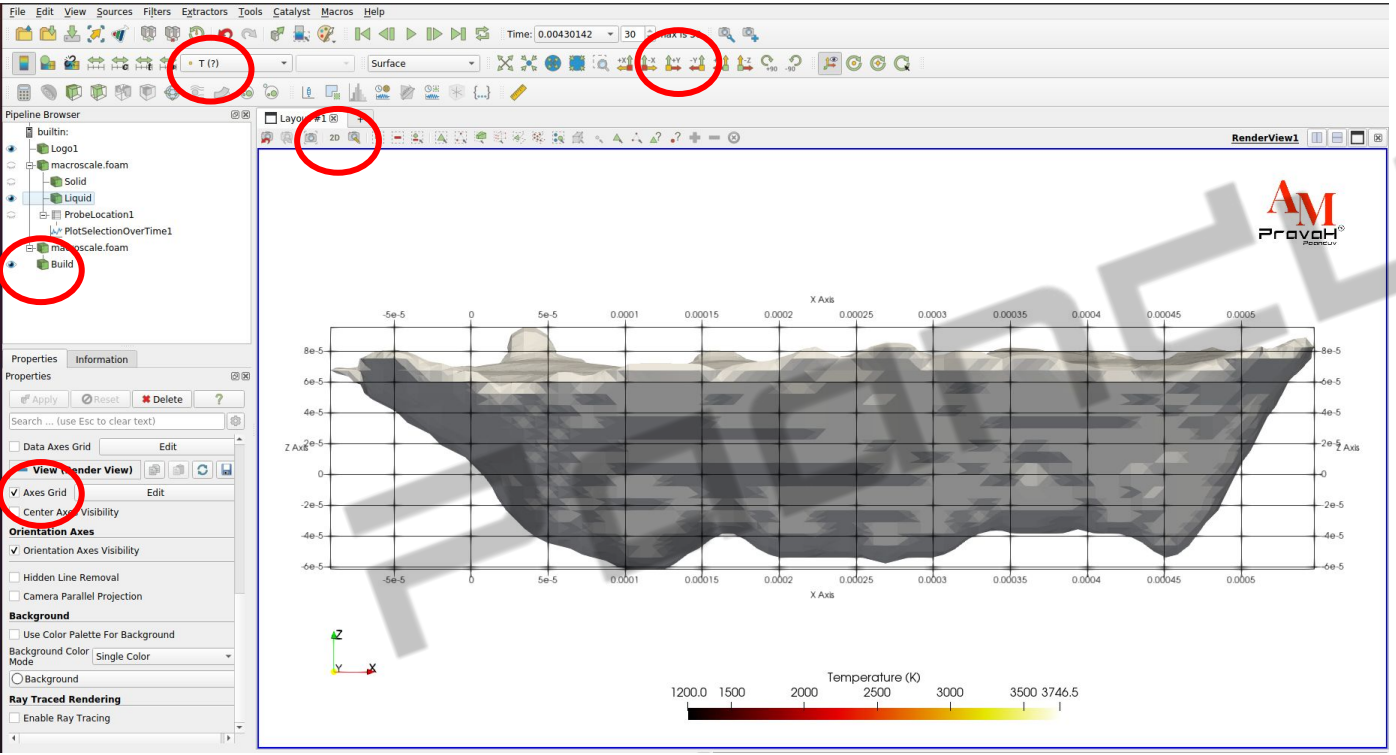
- The clip tool is present in the common toolbox.
- Plane parameters allows selection of orthogonal planes.
- Combination of views and sections allow spatial and temporal growth of microstructure.

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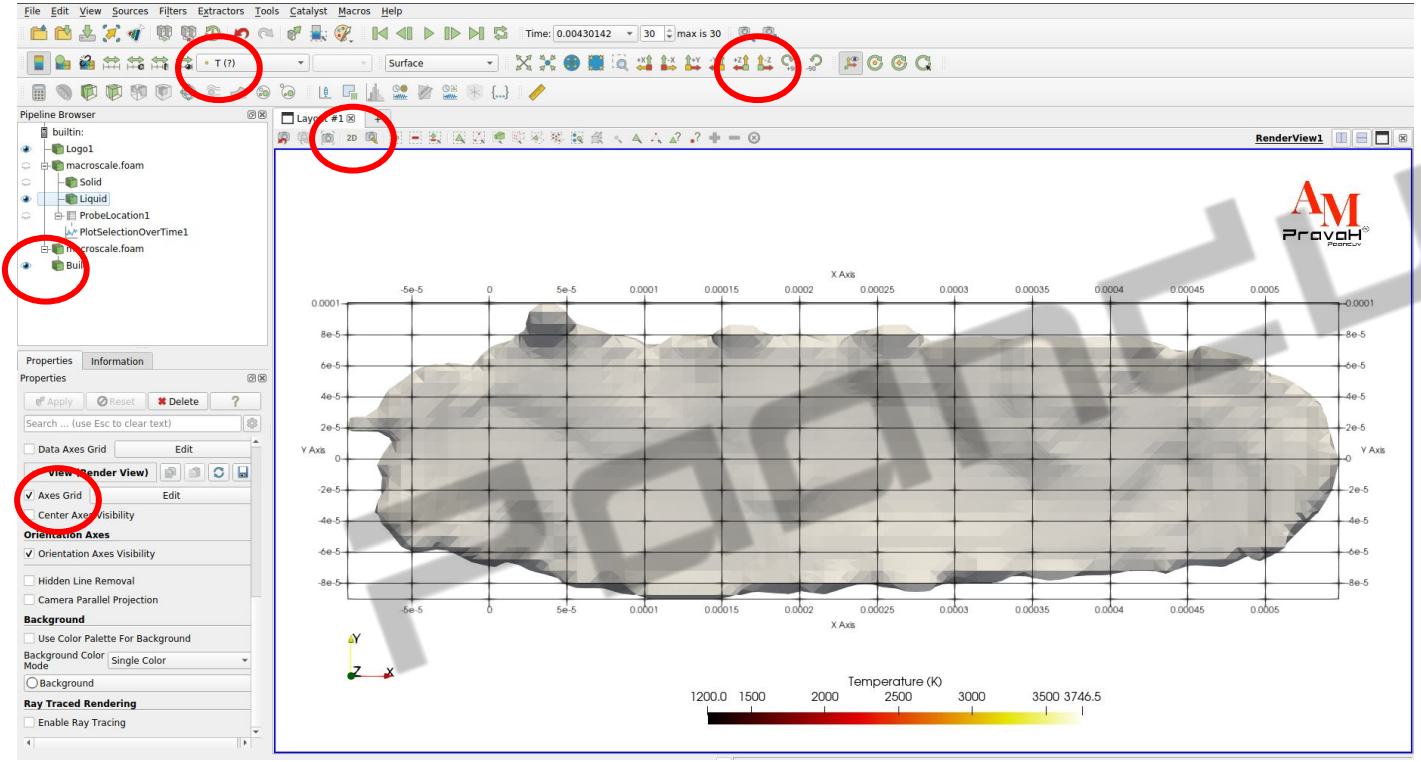
- In pipeline browser choose “Liquid” and “Build”
- In properties choose “Axis Grid” and in “Edit” choose “Cull Front” and “Show Grid”
- Choose “T” from toolbar
- Run time step from “0” to 30 to see meltpool formation and finally build formation

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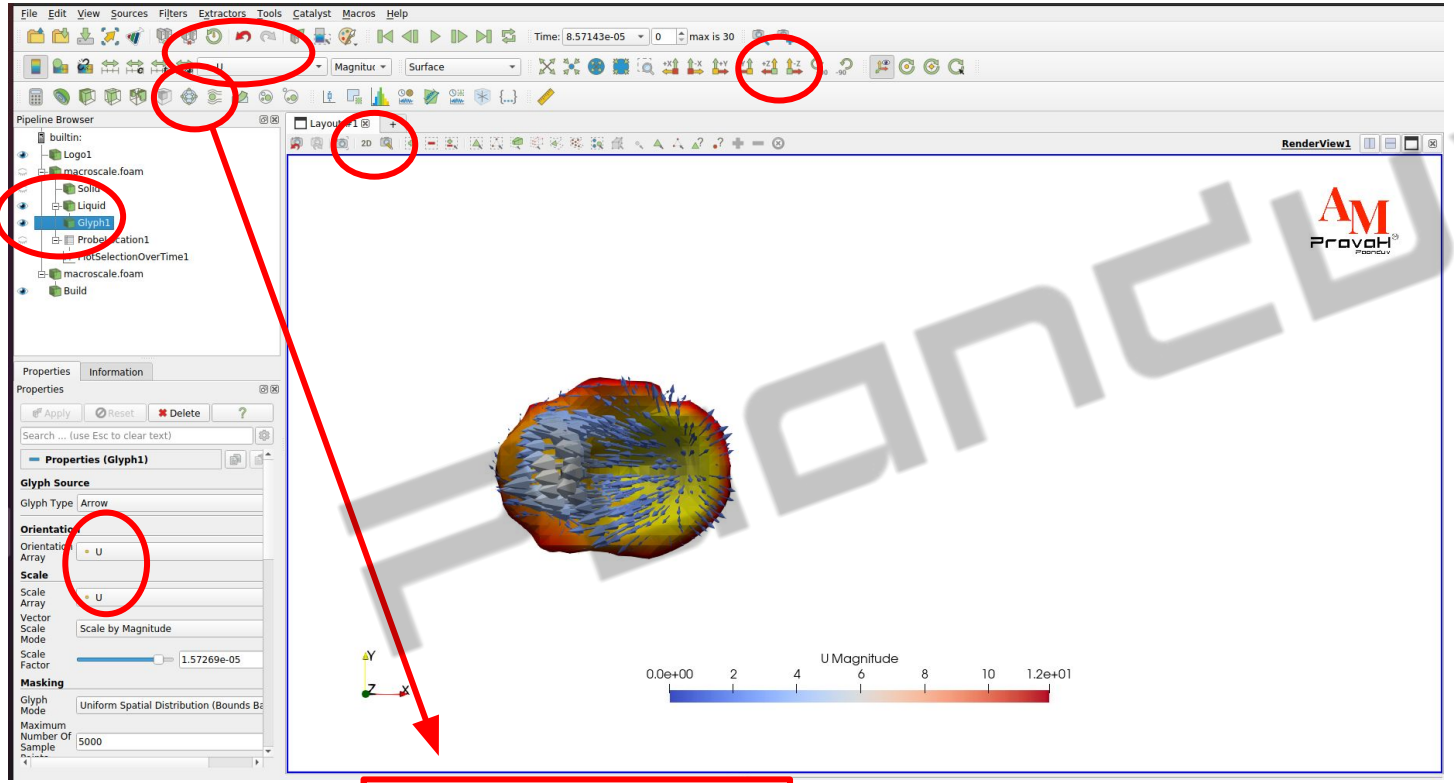
- Choose +Y direction view
- In properties choose “Axis Grid” and in “Edit” choose “Cull Frontface” and “Show Grid”
- Make 2D view and last time step view
- See where build height get constant, from figure it is “0.00028 m” in X-axis.
- Use “Ruler” to measure height by selecting two points at X=0.00028 m

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- Choose +Z direction view
- In properties choose “Axis Grid” and in “Edit” choose “Cull Frontface” and “Show Grid”
- Make 2D view and last time step view
- See where build height get constant, from figure it is “0.00028 m” in X-axis.
- Use “Ruler” to measure width by selecting two points at X=0.00028 m

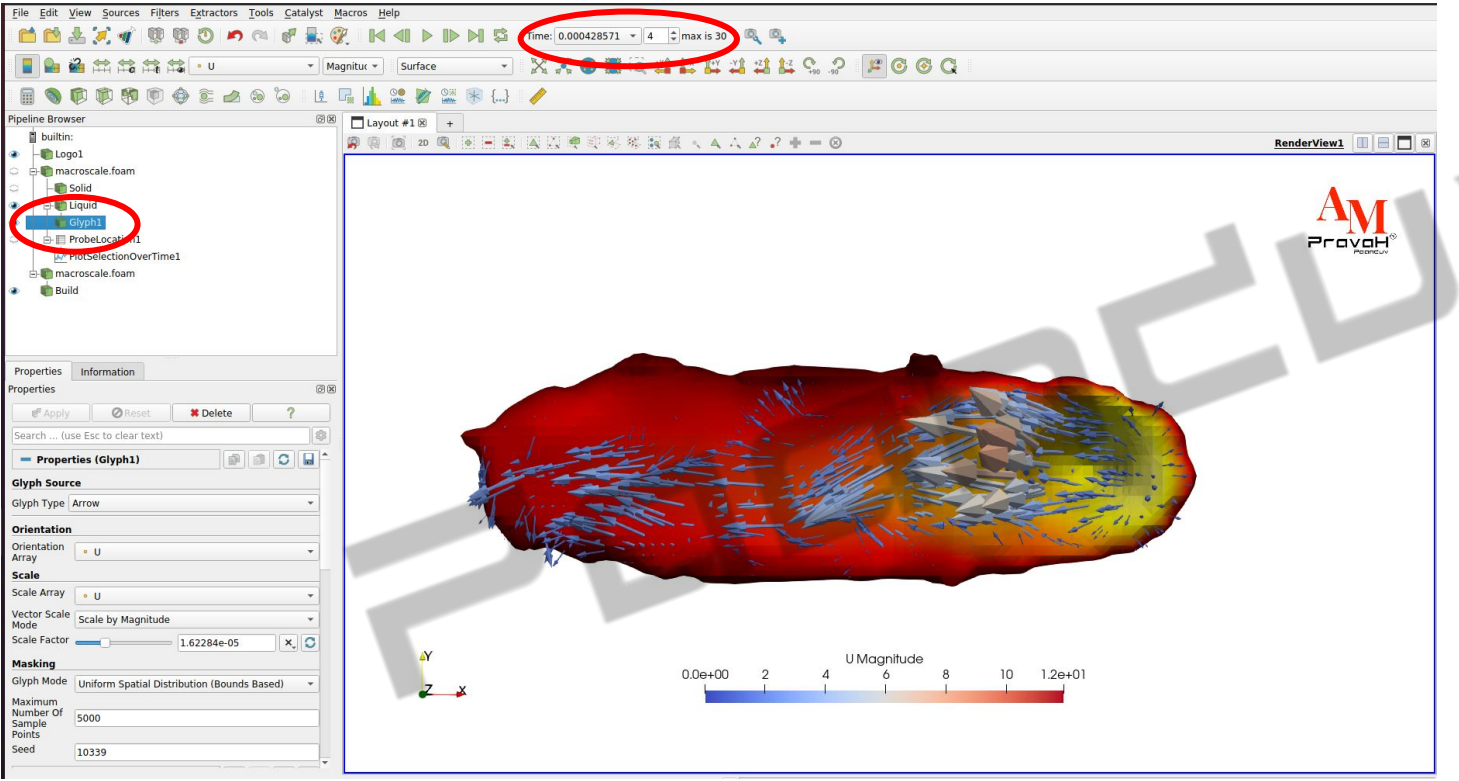
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Glyph

- Choose +Z direction view
- Make 2D view and first time step view
- Initially melt pool starts and downstream molten metal keep solidifying as laser moves +X direction.
- Use “Glyph” to show the Marangoni flow.
- In properties put “Orientation” and “Scale” both as “U”

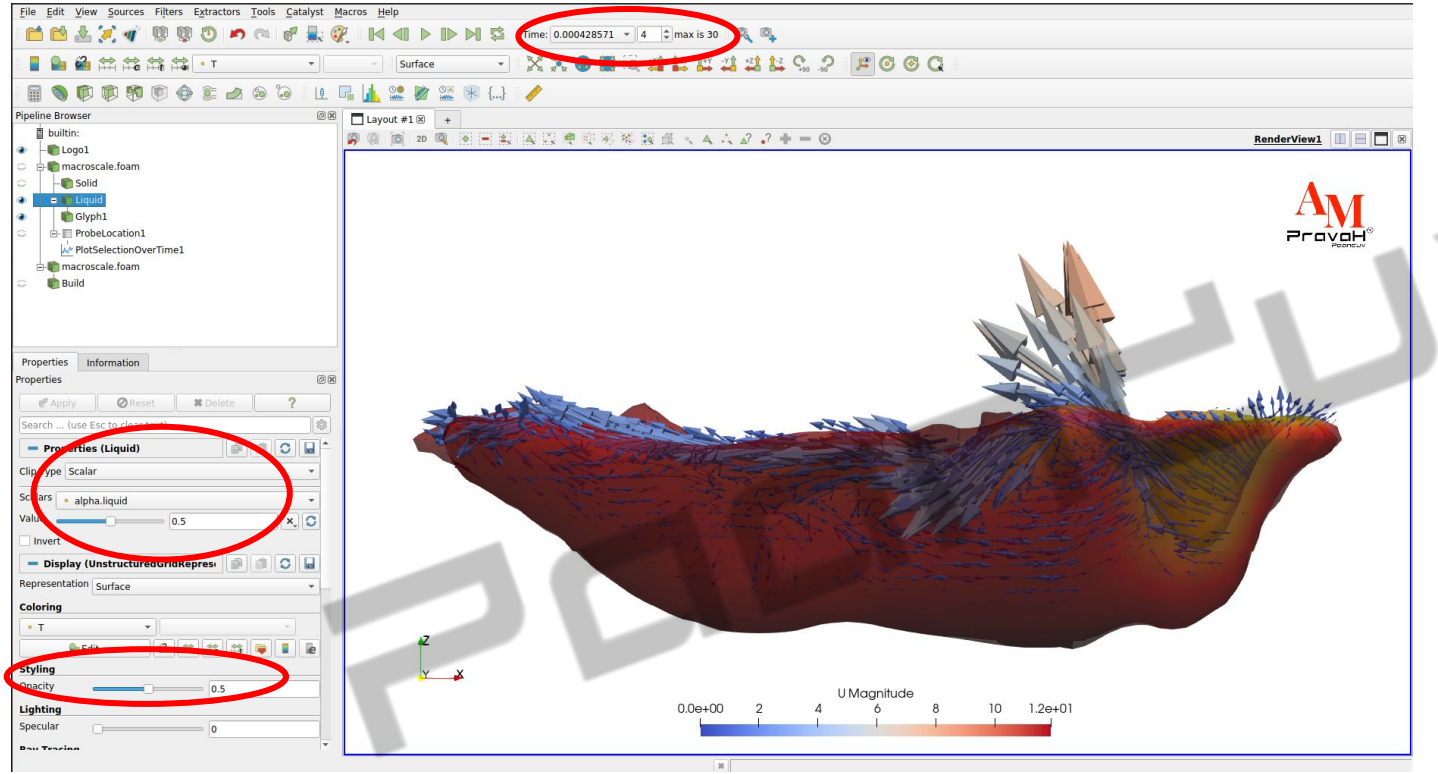
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Top View

- Choose +Z direction view
- Make 3D view
- Initially melt pool starts and downstream molten metal keep solidifying as laser moves +X direction.
- Melt pool is visible as “Liquid” before molten metal solidifies
- Use “Glyph” to show the Marangoni flow.
- In properties put “Orientation” and “Scale” both as “U”

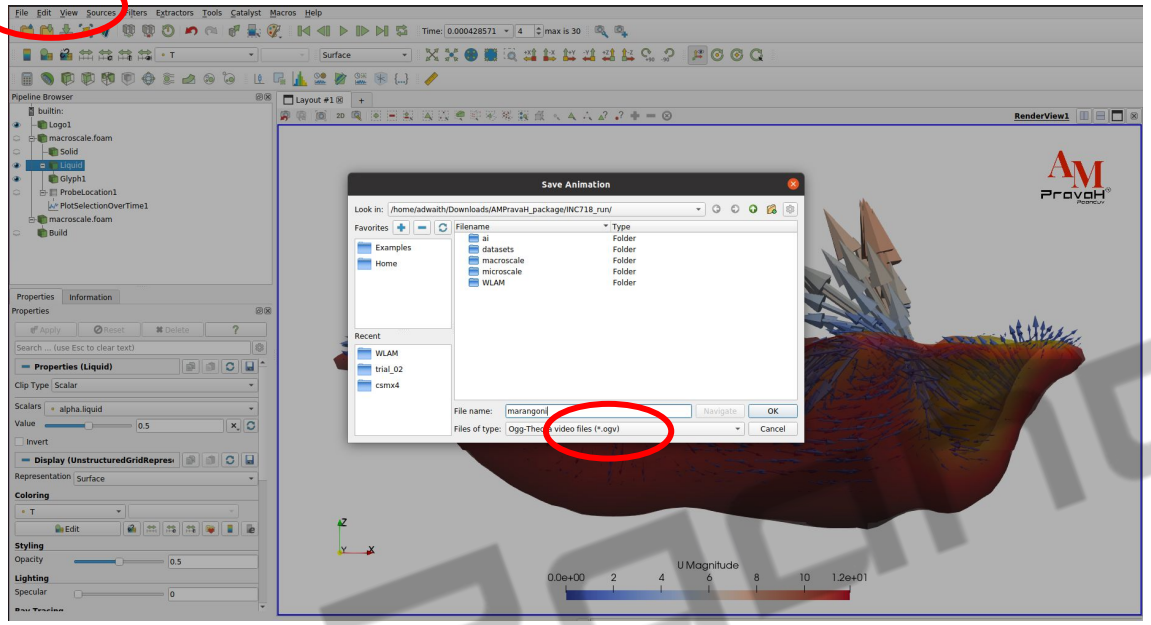
All rights reserved. Paanduv ®



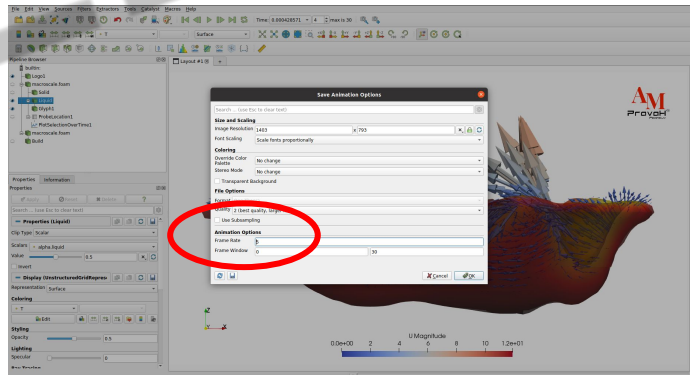
Side View

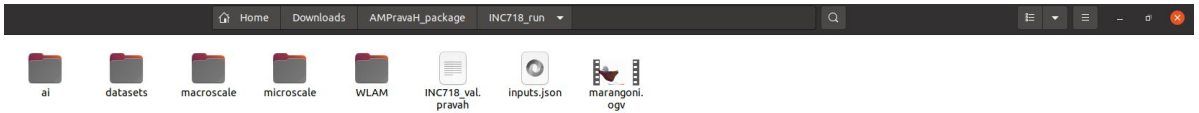
- Choose +Y direction view
- Make 3D view
- Initially melt pool starts and downstream molten metal keep solidifying as laser moves +X direction.
- Make “liquid” opacity to “0.5” to view Marangoni flow inside the liquid also
- Use “Glyph” to show the Marangoni flow.
- In properties put “Orientation” and “Scale” both as “U”

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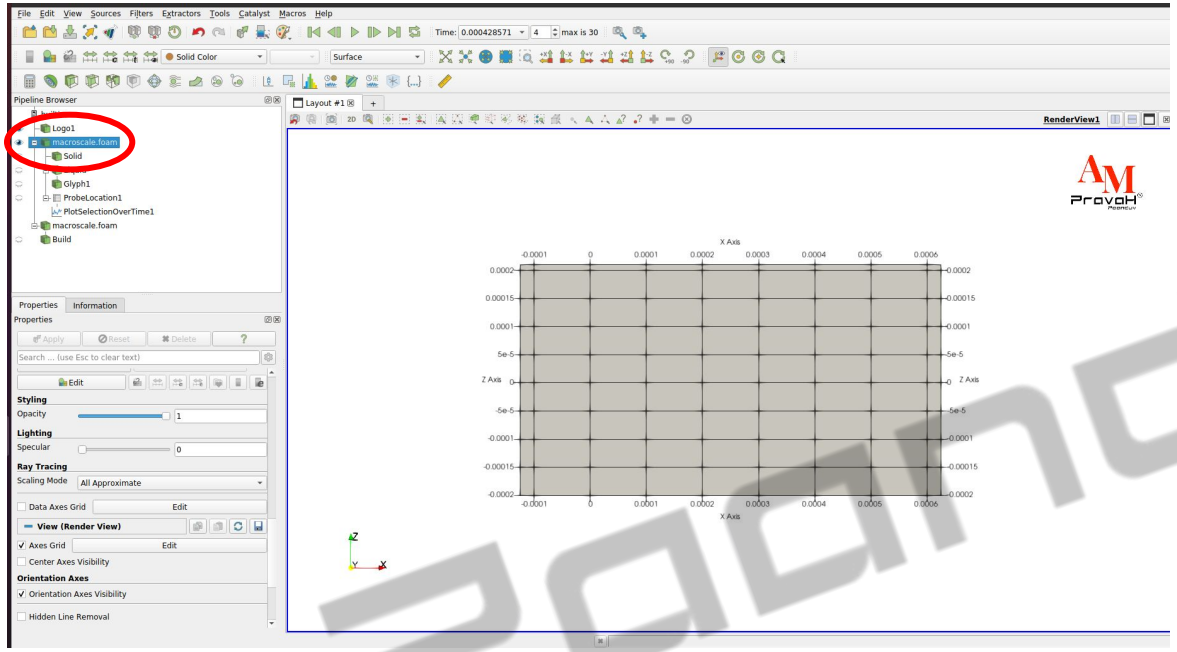


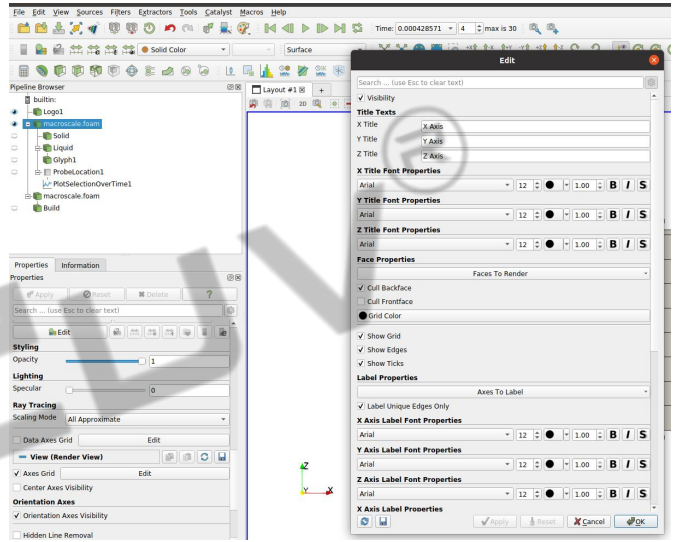
- Save animation from dropdown of “File in first tool bar
- Choose “.ogv” format and save at desired location with frame rate “5”. Higher the frame rate better quality of the animation.





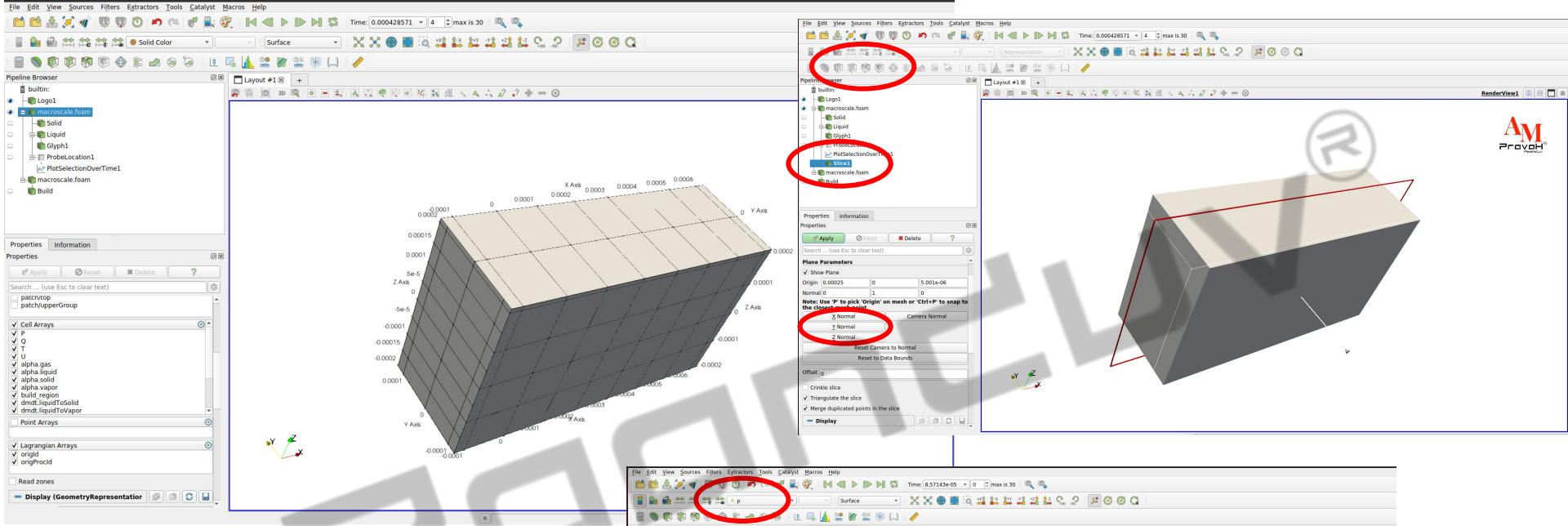
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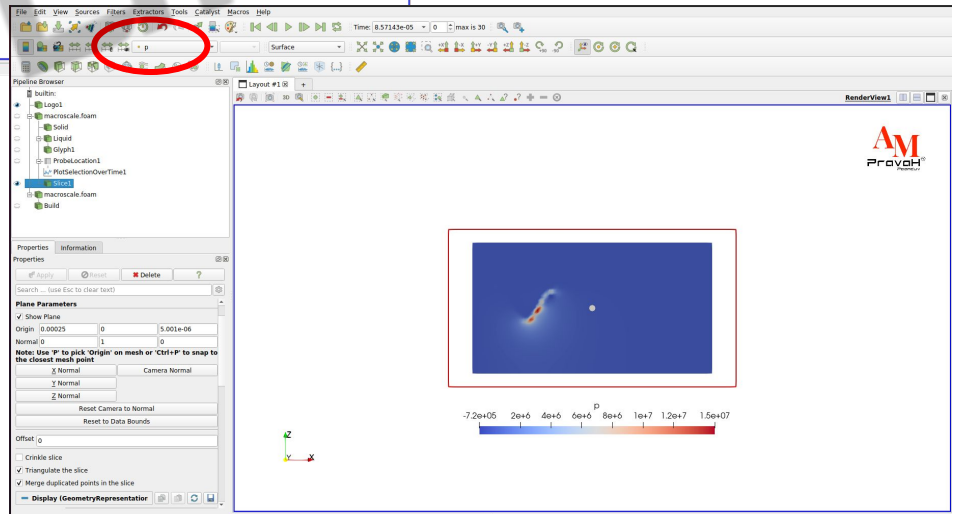


- Choose microscale.foam, ON axis-grid

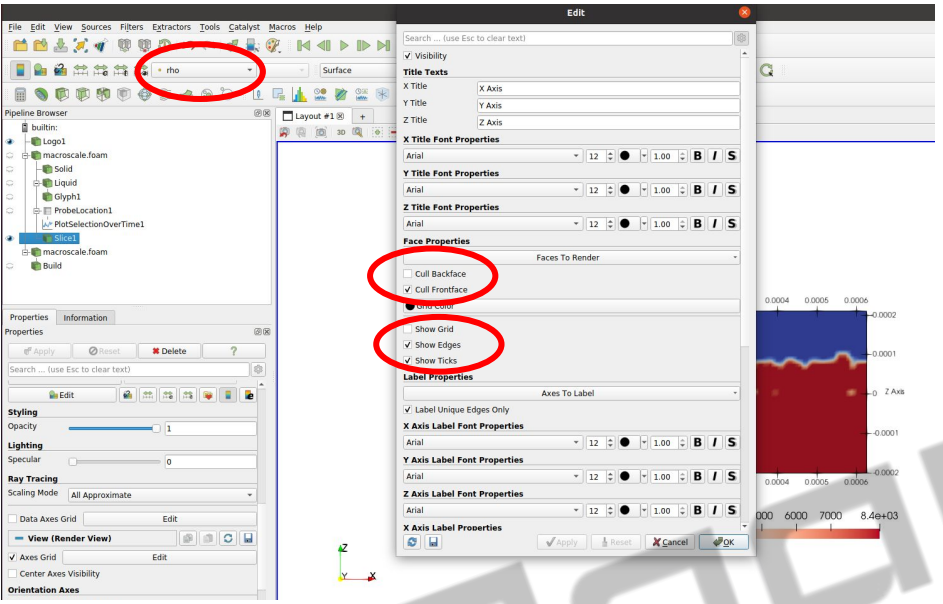
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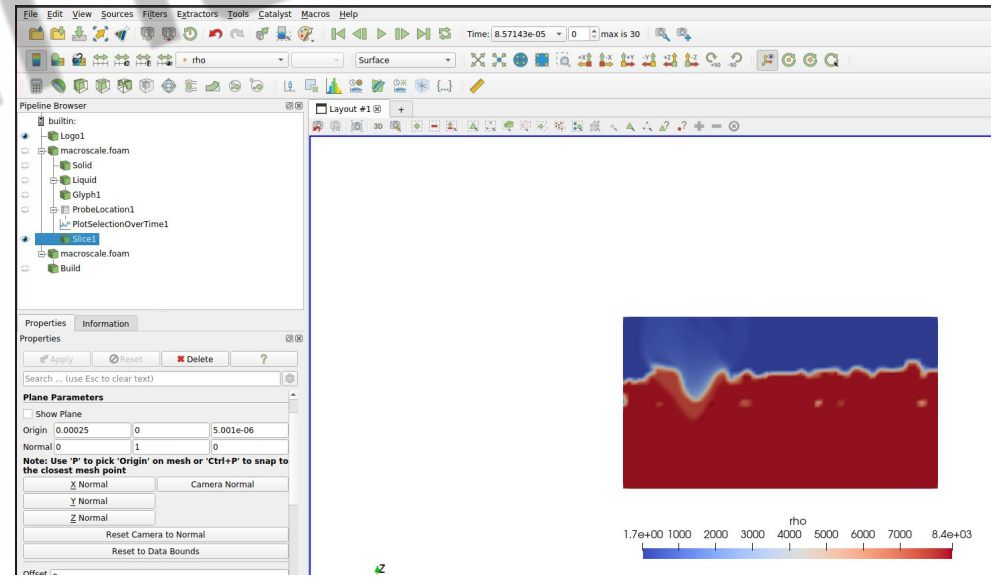


- Make “slice” in Y normal direction, tilt it to see 3D isometric view, and choose property “p” to see pressure distribution



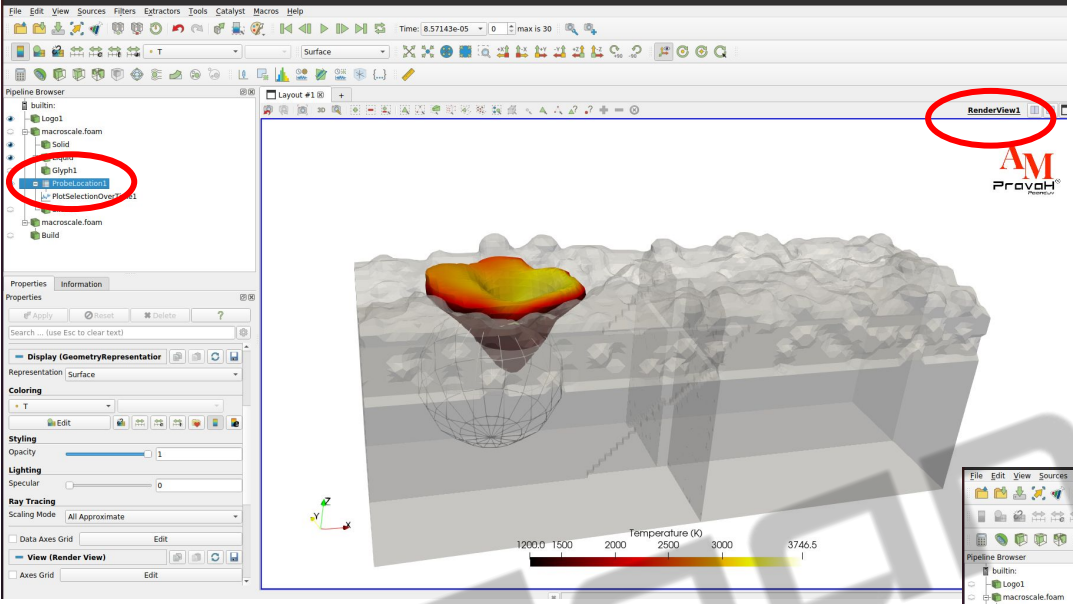
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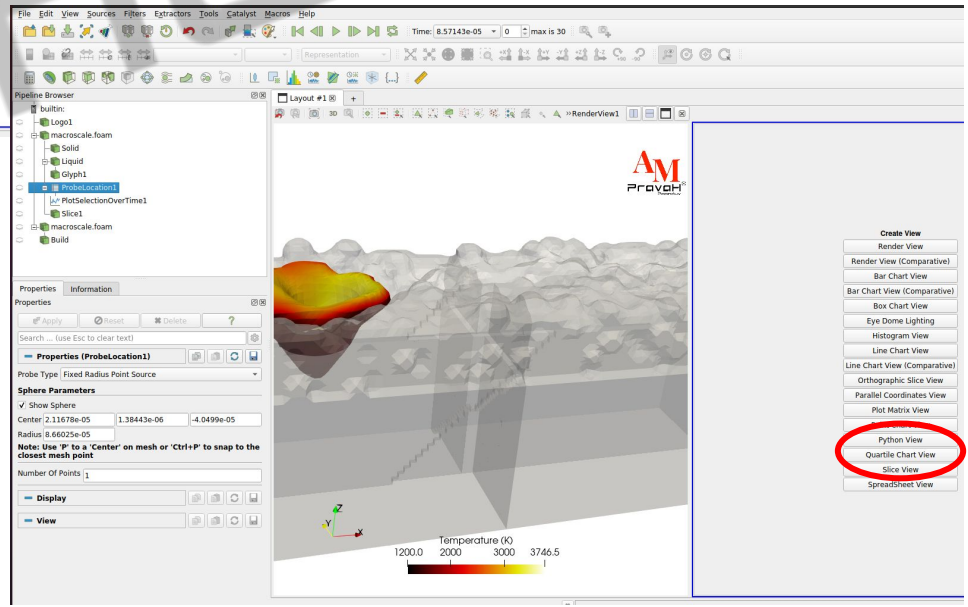




- Run “time step” and make animation and save the file.
- It shows density variation across the melt pool dynamics in the liquid, solid and vapour phase

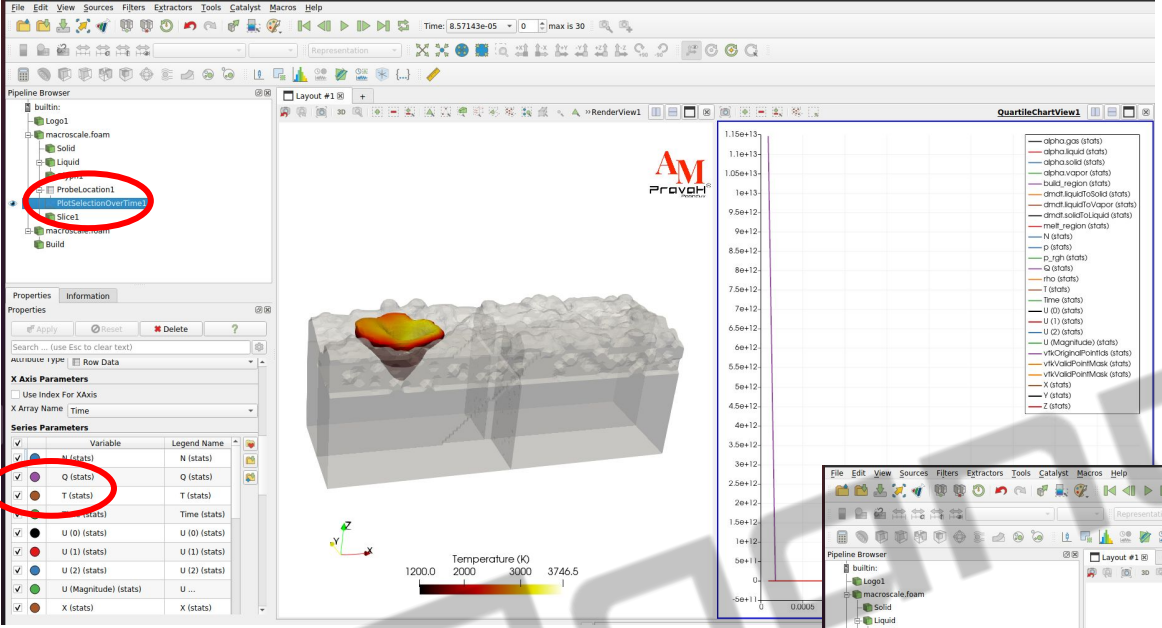
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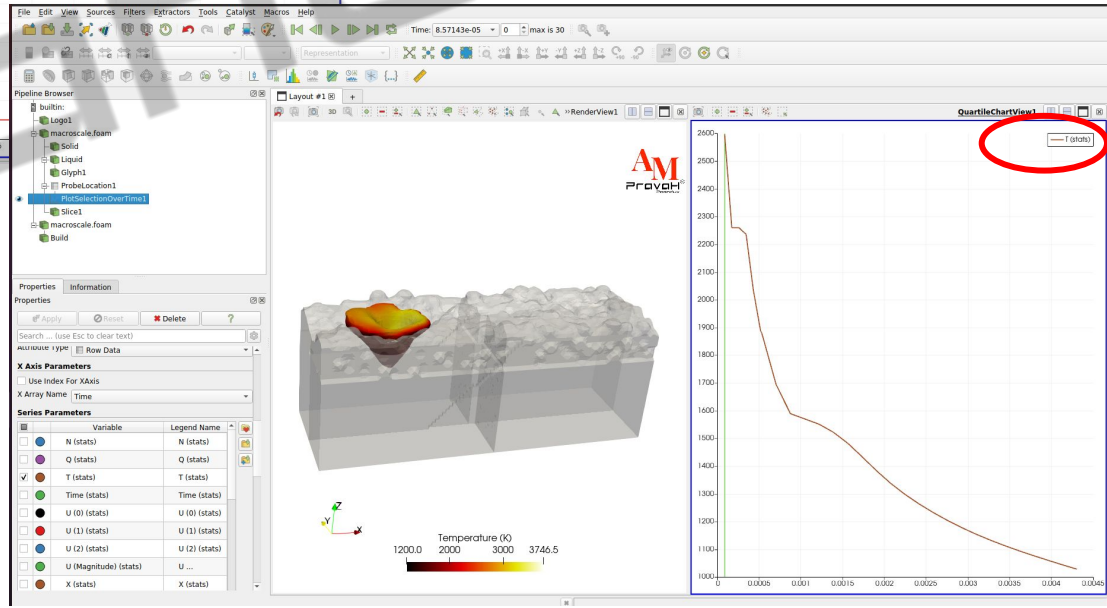




- Choose “ProbeLocation1” from “Pipeline Browser” and set it as default location
- Choose “RenderView 1 and choose “Quartile Chart”.

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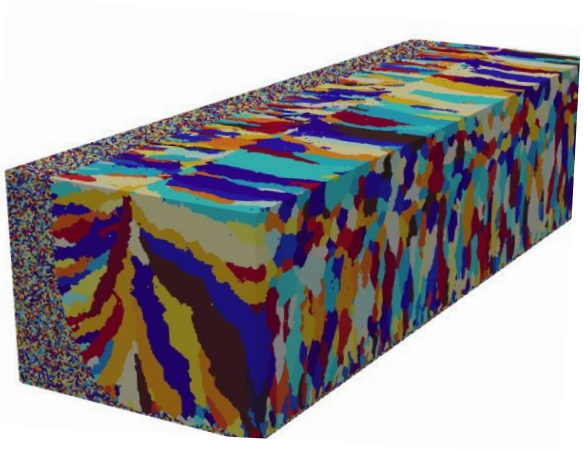




- Select “Quartile Chart” area and choose PlotSelectionOverTime1”
- Choose “Temperature” only from “Series Parameters
- It will give “Temperature variation over the probe location with time

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Microstructure prediction using AM PravaH® microstructure module



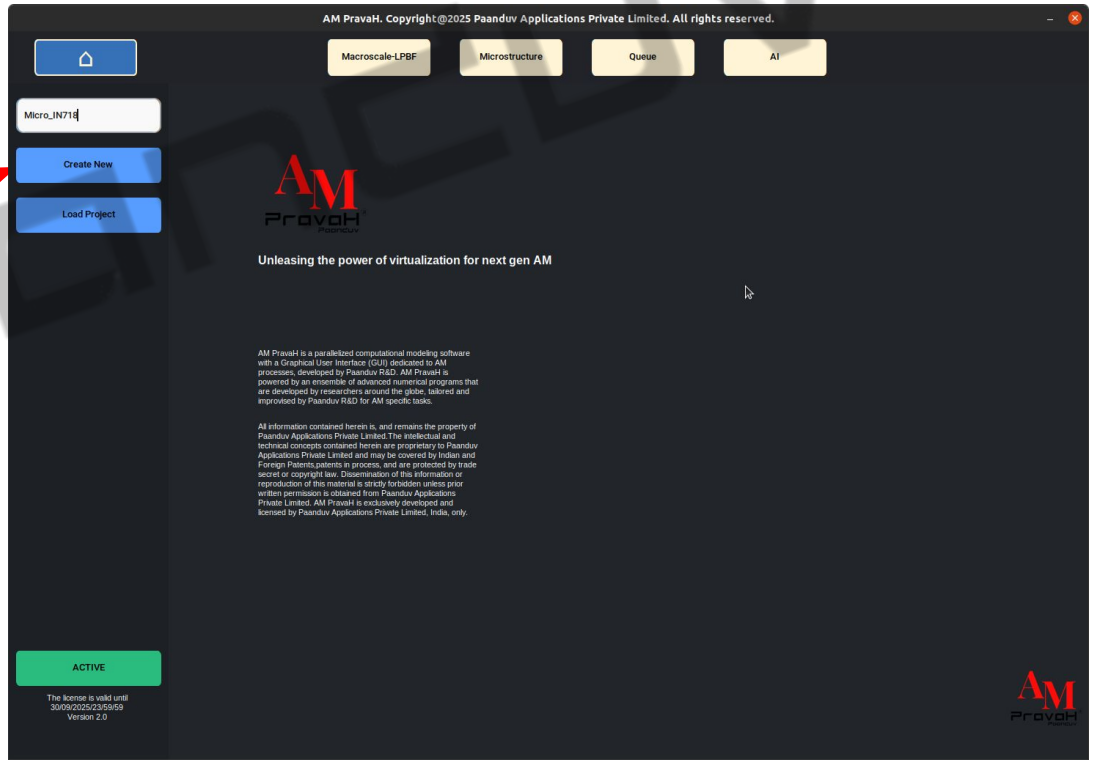
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Hands-on-Orientation on the microstructure module of AM PravaH® 2.0 Graphical User Interface (GUI)

- Pick a desired project file name,

  Let's say Micro_IN718 in our case
- Click on the **Create New** tab below
- Save it to a desired location

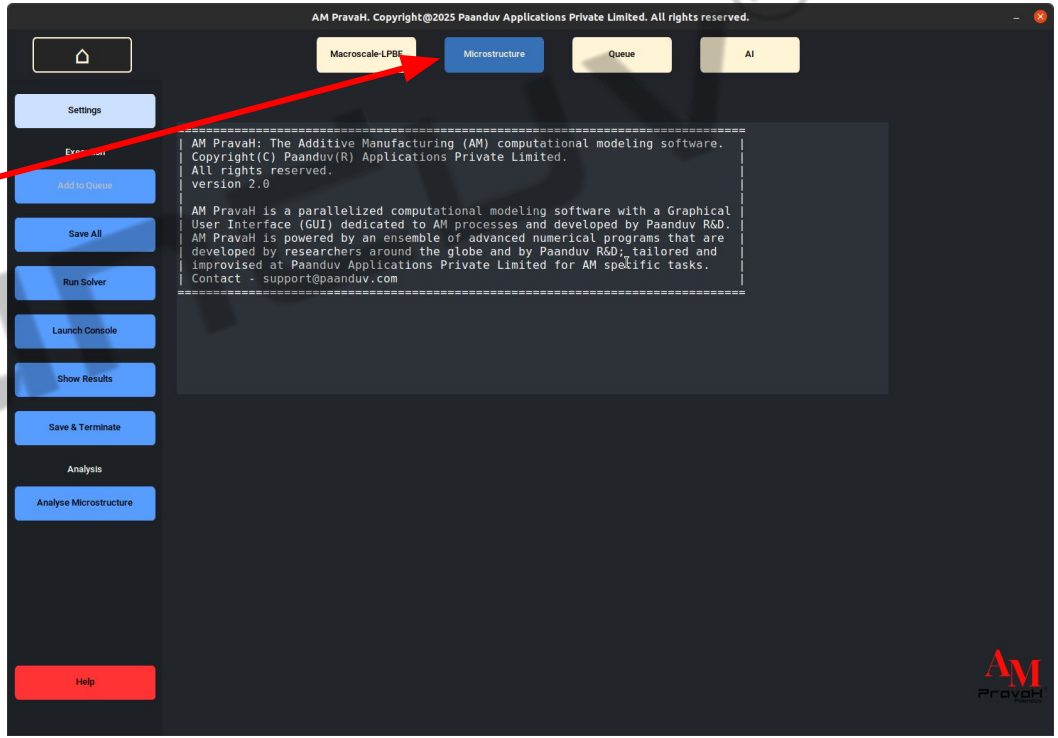


If the file is already created,

Go to **Load Project** tab and open the project (**filename.pravah**) file.

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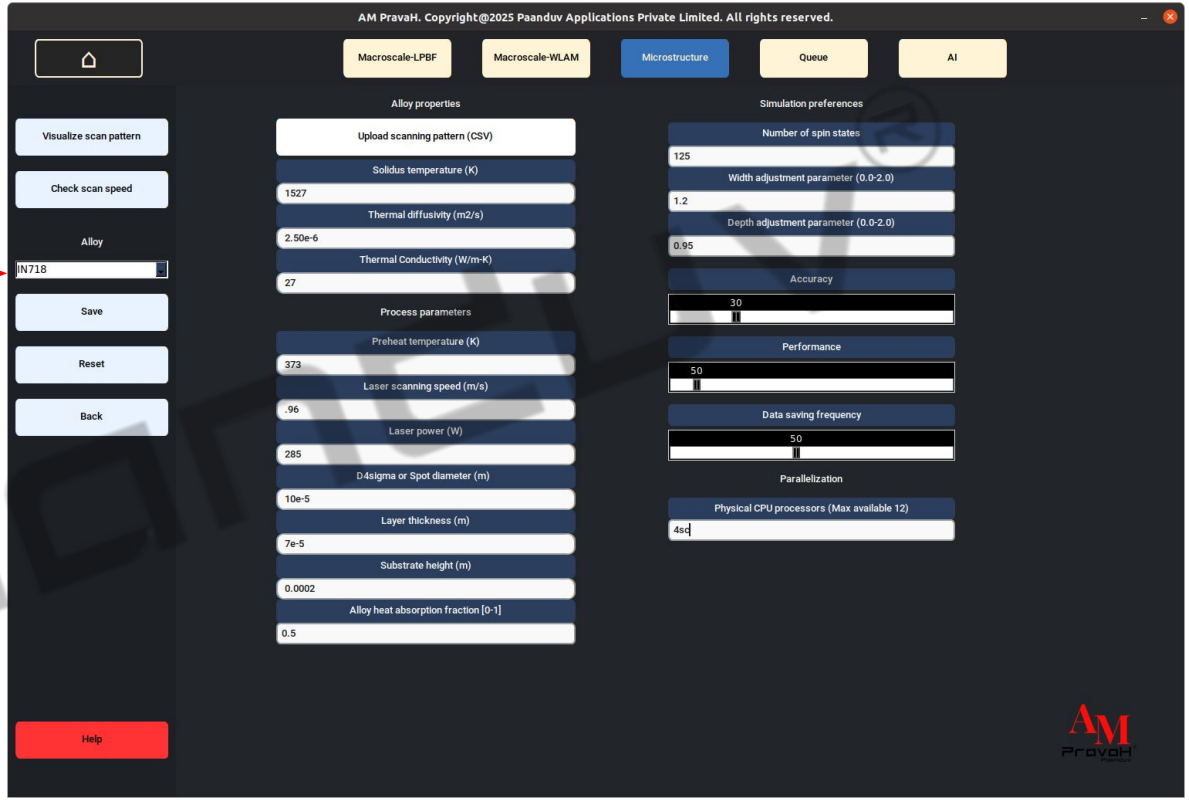
- Opent the AM PravaH GUI
- Choose the **Microstructure** tab
- Select **Settings** tab from the left panel to explore further



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In the Settings tab

- Select the desired alloy from the left panel
- A drop-down list of different alloys will appear
- Scroll down and select the alloy, in our case choose IN 718.
- Input the alloy properties, import scan pattern file, process parameters, simulation parameters and parallelization scheme
- The fidelity of the simulation is controlled by the sliding the Accuracy, and Performance.



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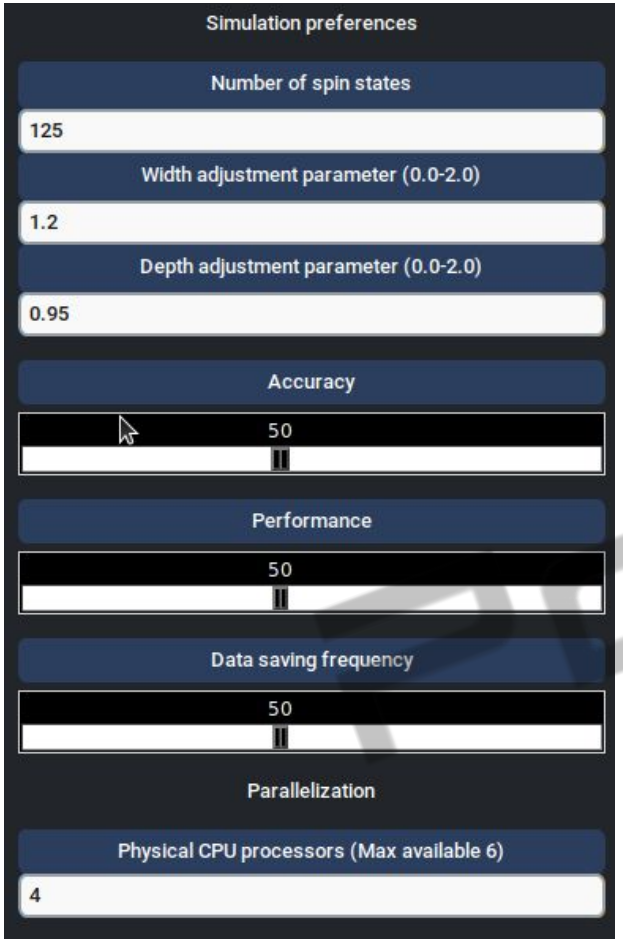
Thermophysical properties are given as inputs:

- Solidus temperature (in K),
- Thermal Diffusivity (in m^2/s)
- Thermal conductivity (in W/m-K) (Based on literature or CALPHAD)

Input process parameters:

- Scan strategy or pattern file (import as a .csv file)
- Bed preheat temperature (in K)
- Laser Scan speed (in m/s)
- Laser power (in W)
- Spot diameter (D4 sigma) (in m)
- Powder layer thickness (in m)
- Substrate height (in m)
- Heat absorption fraction is an iterative parameter.

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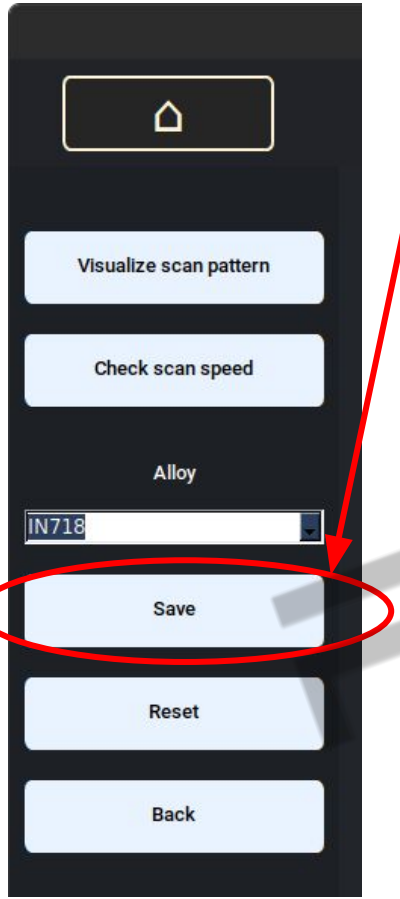


- Simulation preferences include defining the grain IDs (unique grain identifiers). Generally, 125 unique spin states are defined as defaults.
- Width and depth parameters (Ref. to the simulation)

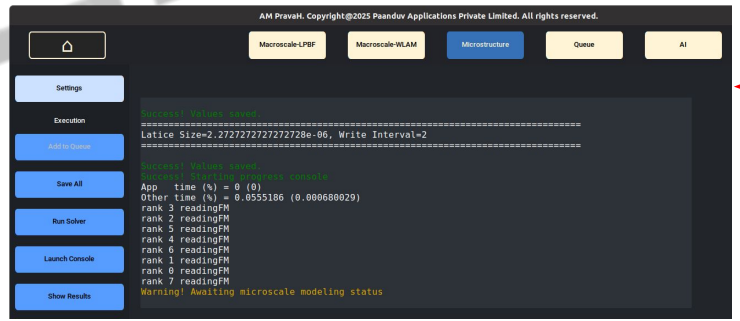
Processing efficacy: (Controls computation time)

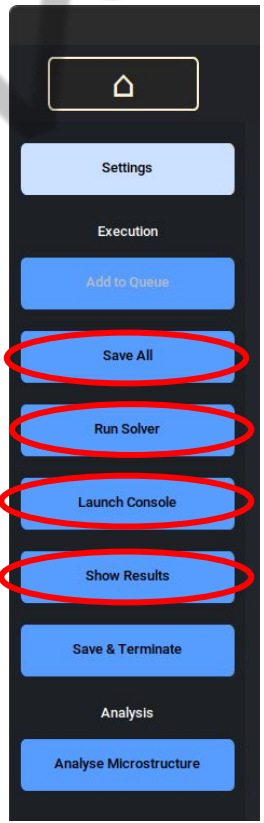
- Accuracy and performance determines resolution in the evolved structure. Computation cell size gets smaller at higher accuracies.
- Data saving frequency determines the time step.
- Parallelisation divides the domain in multiple processor cores.

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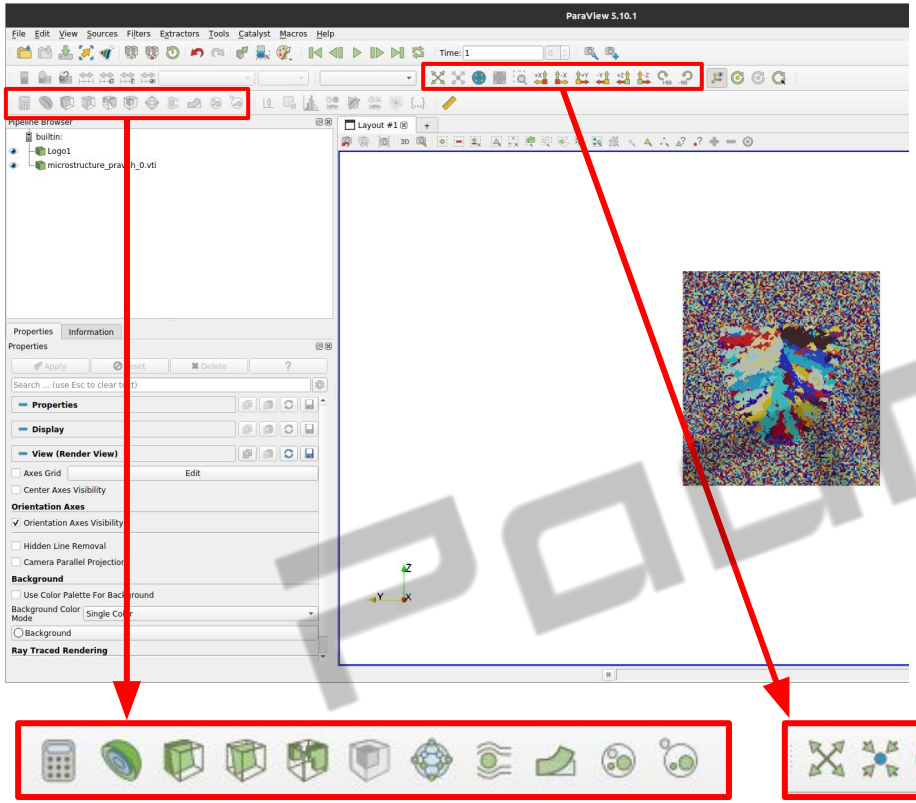


- Save all the parameters & preferences
- Go back to the project window, and click on the **Save All** tab in the left panel, to finalize the settings for a particular project.
- Click on the **Run solver** tab, to run simulations
- Click on the **Launch Console** tab. It helps in checking the status of the simulation.
- Press on **Show Results** to go to Paraview.





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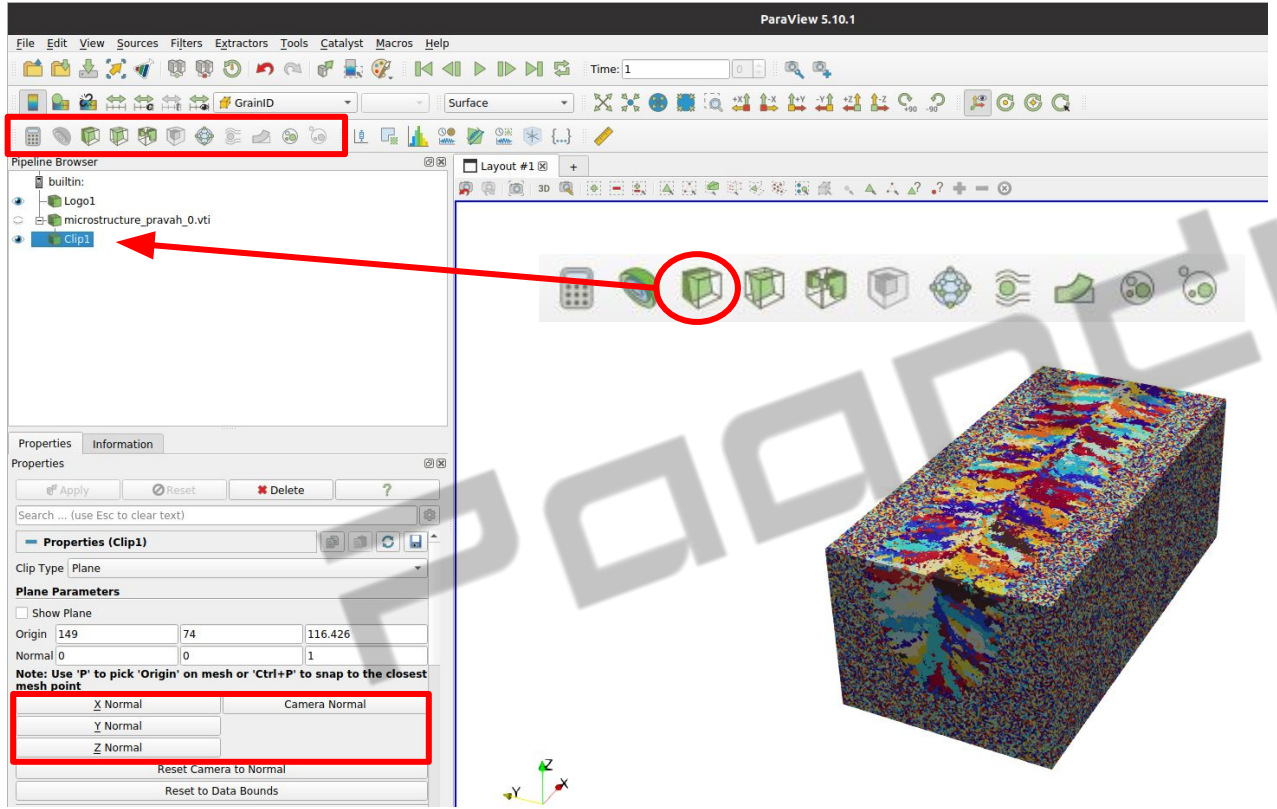
- The Paraview creates and renders the simulation
- Use the different features like clip & slice to view different sections of the melt pool
- Important tool boxes include, common tool box, camera control, etc.
- Different views will be listed in the Pipeline Browser (located at the left panel) contains all the views and clips.

Common toolbox

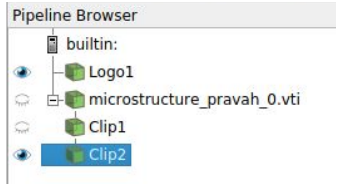


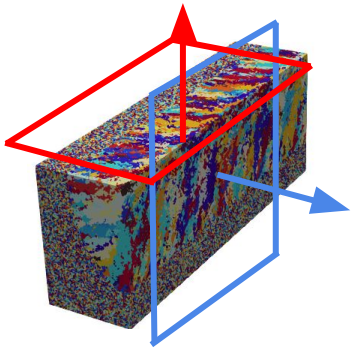
Camera control toolbox

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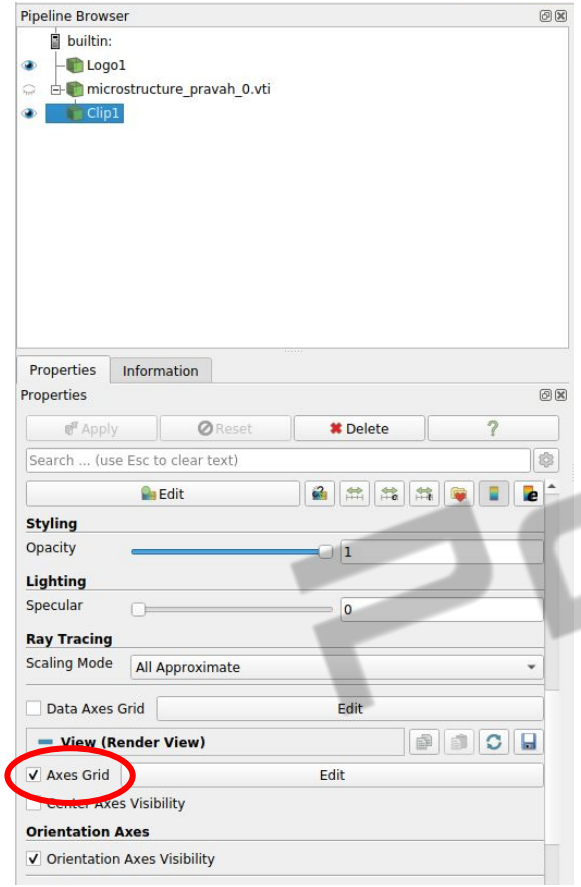


- The clip tool is present in the common toolbox.
- Plane parameters allows selection of orthogonal planes.
- Combination of views and sections allow spatial and temporal growth of microstructure.



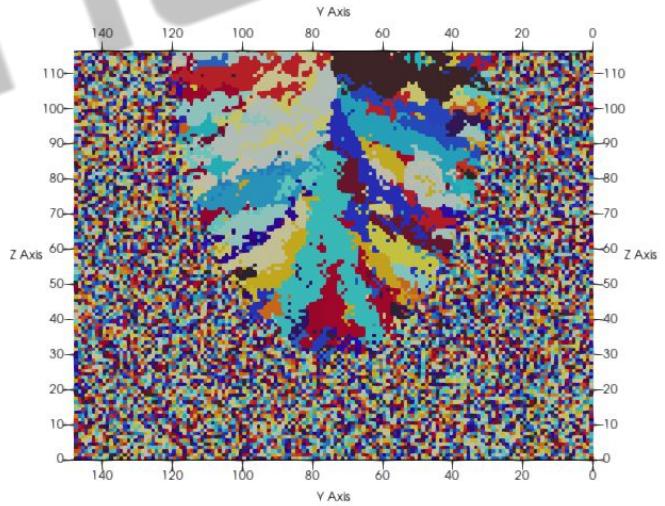


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Putting scale for the views

- Trim the undesired sections and keep the view in 2D
- Select the **Axes Grid** from the left panel to represent the scales
- The units of the grid can be edited by pressing the Edit button adjacent to it.

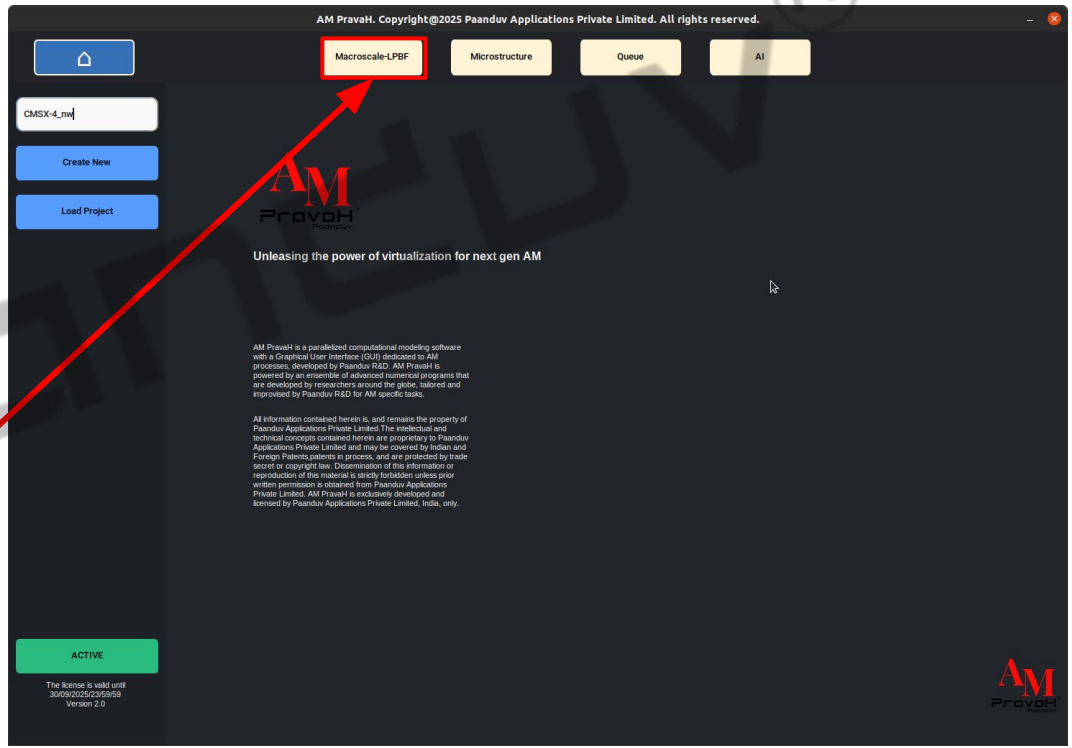


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Adding a new alloy in the database

- Pick a desired project file name,
  Let's say CMSX-4_nw in our case
- Click on the **Create New** tab below,
  in the left panel
- Save it to a desired location

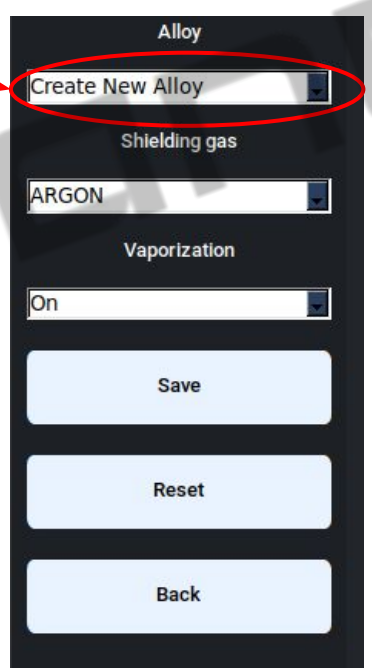
Go to **Macroscale-LPBF** tab on top row, and click on it

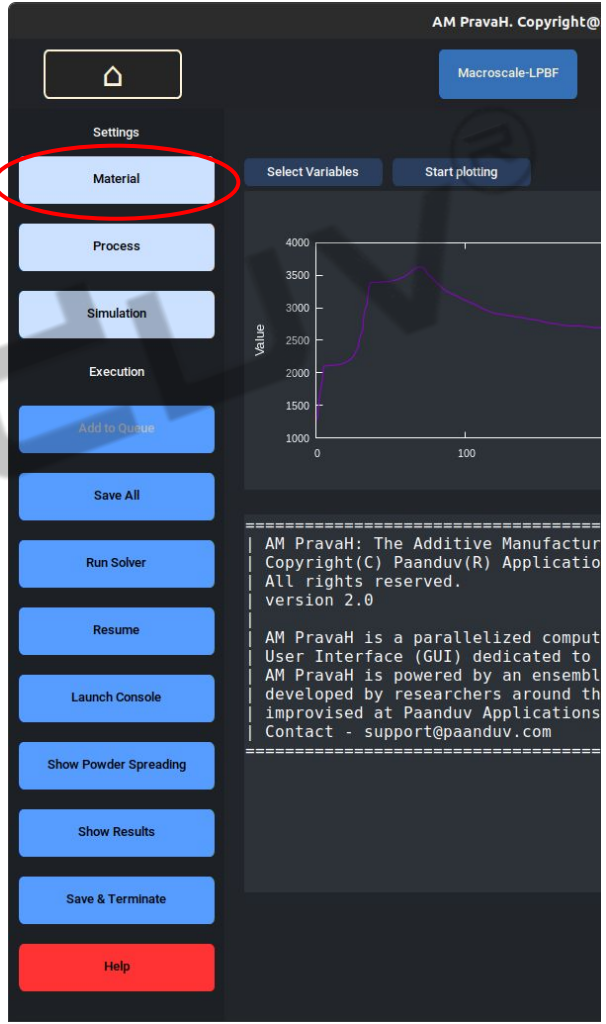


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In the Macroscale-LPBF window:

- Click on the **Material** in the left panel
- Inside Material window, select **Create New Alloy** from the drop down alloy list in the **Alloy section**
- Choose the **Shielding gas** (if any)
- The user can turn on and off the **Vaporization** .

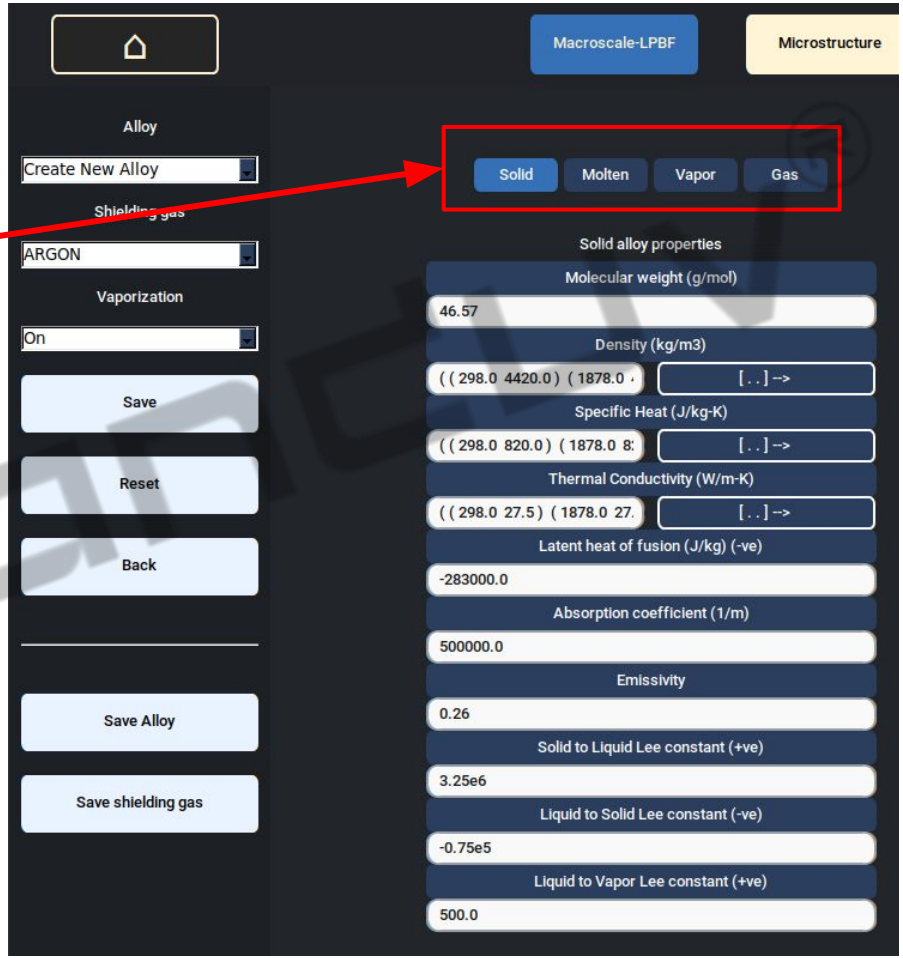




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Choosing the **Create New Alloy** option enables the user to add the thermophysical properties as inputs for the **Solid, Molten, Vapour** and **Gas** phase

Note: The thermophysical properties used for the simulation are temperature dependent and can be obtained either from the literature or using CALPHAD.



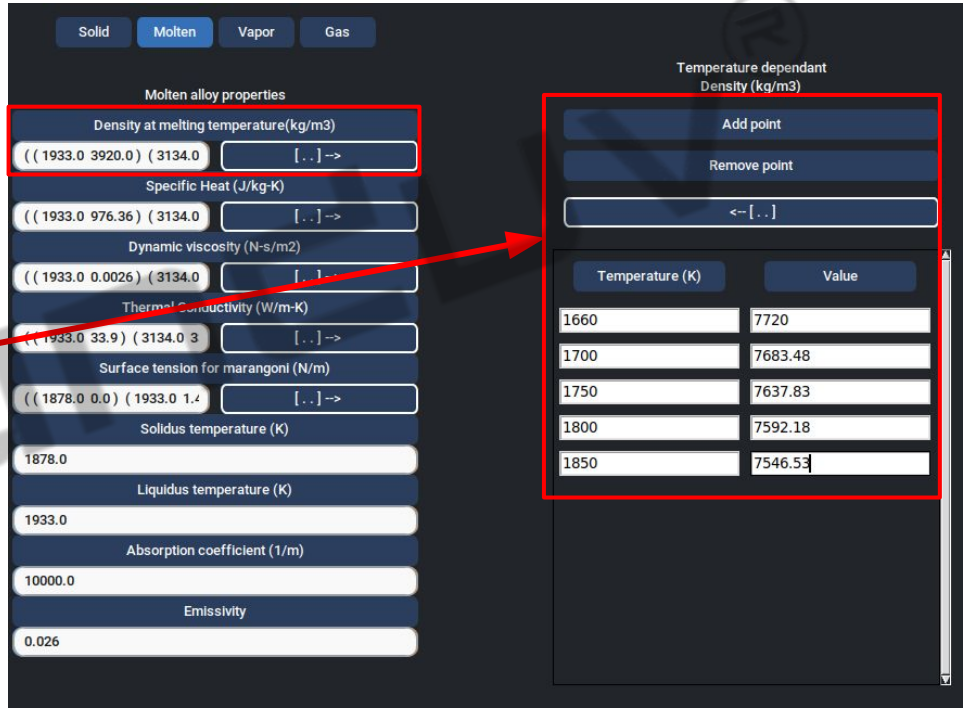
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For demonstration purposes, the temperature dependent values of density in the **Molten** phase will be added for CMSX-4 alloy.

Similarly, fill the other parameters like Dynamic viscosity, thermal conductivity, the respective temperatures for all the phases.

Save the values by pressing the **Save** tab.

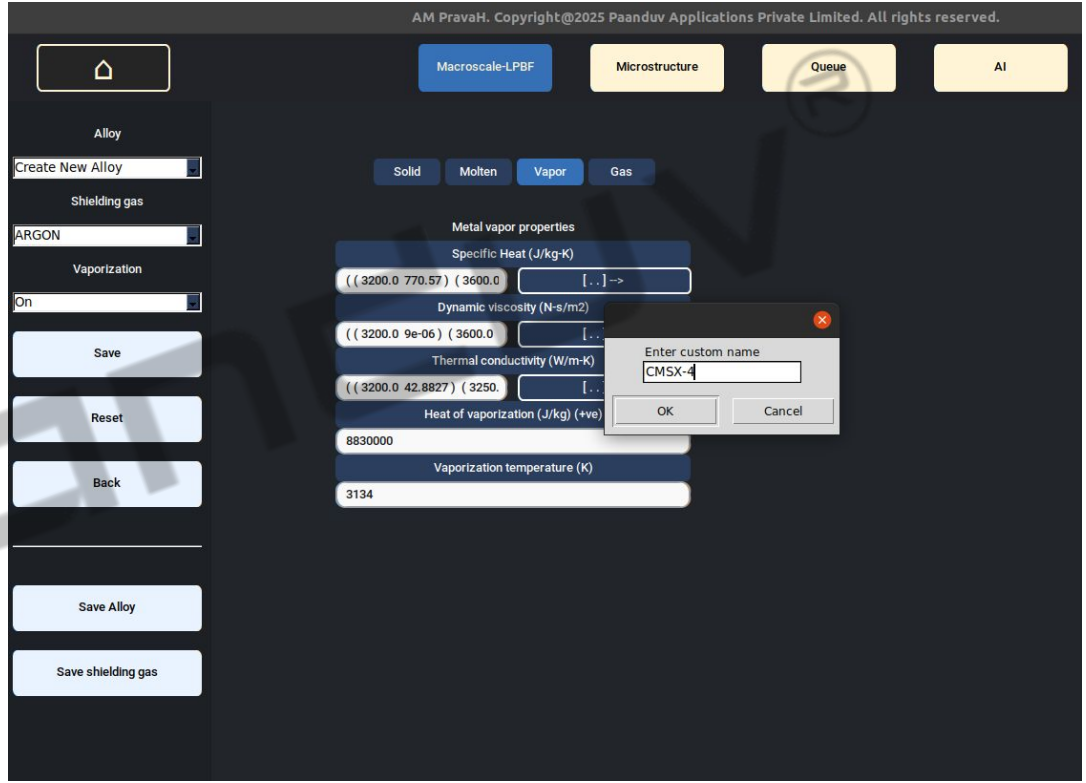
Density	
Temp (K)	Kg/m3
1660	7720
1700	7683.48
1750	7637.83
1800	7592.18
1850	7546.53
1890	7510.01
2900	6587.88



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Click on the **Save Alloy** tab in the left panel and enter the name (CMSX-4, in this case) of the specific alloy to add it on the drop-down list of the Alloy section.

Go to **Microstructure** tab and choose CMSX-4 (newly added on the drop-down list) and follow the steps mentioned above to simulate the met pool dynamics and microstructure.



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Upload your queries on Cfd.online:

https://www.cfd-online.com/Forums/main/260598-substrate-temperature-rise-multi-layer-simulation-using-ampravah.html

DEMO REQUEST FORM

Thank you.

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